

Crafting clarity: Leveraging large language models to decode consumer reviews

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ABSTRACT

Large Language Models (LLMs) have emerged as powerful tools for understanding consumer perceptions and extracting insights from unstructured textual data. This study investigates the effectiveness of LLMs in comprehending consumer opinions, particularly in service industries. We fine-tuned four prominent LLMs—Falcon-7B, MPT-7B, GPT-2, and BERT—using 1,031,478 consumer reviews and assessed their ability to identify topics, emotions, and sentiments. Our results indicate that Falcon-7B excels in accuracy and reliability for complex tasks. This study is the first, to our knowledge, to fine-tune Large Language Models (LLMs) specifically for consumer data, showcasing the efficacy of attention mechanisms in extracting valuable insights. Our findings offer strategic decision-making insights for the service industry and underscore the transformative potential of LLMs in business intelligence from consumer feedback.

1. Introduction

In the evolving service sector, the importance of online consumer reviews cannot be overstated (Cheung and Lee, 2012; Filieri, 2015). As the internet and mobile technologies advance, online reviews have become critical for consumers to evaluate the quality of products and services before purchases, significantly shaping consumer perceptions and decision-making (Hu et al., 2008; Sparks and Browning, 2011; Wu and Chang, 2020). Rooted in real customer experiences, these reviews are seen as genuine and impartial, thereby influencing purchase decisions (Sparks and Browning, 2011). Traditionally, consumers faced difficulties in assessing service quality without firsthand experience. However, online review platforms have facilitated broader access to evaluative information, enabling consumers to make better-informed decisions based on the experiences shared by others (Sotiriadis et al., 2013).

Online reviews manifest in various formats, such as ratings and textual feedback, with the latter providing a deeper dive into consumers' feelings and choices. Textual reviews in sectors like tourism highlight service quality, customer contentment, and booking likelihood, aiding potential tourists in making informed choices and reducing purchase risks (Utz et al., 2012; Mudambi and Schuff, 2010). Given the

voluminous nature of online review data, sophisticated technologies such as big data analytics, natural language processing (NLP), and artificial intelligence (AI) are essential for analysis, surpassing traditional manual techniques (Kang et al., 2020; Rajput, 2020). Employing these technological tools enables thorough mining and examination of online review content, offering deeper insights into consumer attitudes and preferences.

Over the past decade, a major advancement in natural language processing (NLP) has been the introduction of transformer architectures by Google scientists, highlighted in the revolutionary work "Attention is All You Need" by Vaswani et al. (2017). This groundbreaking development addresses the disadvantages of traditional neural network architectures like recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, paving the way for models that rely on self-attention mechanisms. This shift has significantly enhanced the computational efficiency of NLP tasks, facilitating the development of large language models (LLMs) with extraordinary capabilities (Devlin et al., 2018; Brown et al., 2020; Kucharyav et al., 2023), capable of handling unstructured, complex textual data such as online customer reviews (Radford et al., 2019). Unlike traditional machine learning methods, LLMs learn directly from raw text, eliminating the need for manually crafted features and specialized domain knowledge, as

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explained by Liu et al. (2019). This approach enables LLMs to discern the finer nuances and semantic interrelations within diverse textual landscapes.

Previous literature in consumer studies predominantly employed Bayesian models, such as Latent Dirichlet Allocation (LDA) and Structural Topic Models (STM), for analyzing consumer data (Blei et al., 2003; Roberts et al., 2013; Kumar et al., 2023; Basu et al., 2024; Luong et al., 2024). However, these frequency-based models often struggle to capture the intricate and nuanced elements of textual data due to the lack of an attention mechanism, a key feature of modern transformer-based Large Language Models (LLMs). This gap has led to a shift towards LLMs, which are recognized for their superior ability to understand and generate text across various domains (Radford et al., 2019).

Transfer learning has become essential for customizing pre-trained Large Language Models (LLMs) for specific domains. In our research, we employed prompt engineering techniques to fine-tune four LLMs—BERT (Bidirectional Encoder Representations from Transformers), Falcon-7B, GPT-2 (Generative Pre-trained Transformer 2), and MPT-7B (Mosaic Pretrained Transformer-7 billion parameter model)—for analyzing consumer reviews (Devlin et al., 2018; Almazrouei et al., 2023; Radford et al., 2019; MosaicML NLP Team, 2023). Traditionally, machine learning algorithms like decision trees, Naïve Bayes, and Random Forest were used in text analytics for their simplicity and ease of interpretation. However, they struggled with capturing semantic relationships and contextual nuances in unstructured textual data (Wang and Wang, 2023; Li et al., 2022). Transfer learning enables LLMs to be fine-tuned with domain-specific data, such as consumer reviews from platforms like TripAdvisor, enhancing their relevance and effectiveness (Devlin et al., 2018; Almazrouei et al., 2023). Fine-tuning is crucial because, while pre-trained LLMs excel at general language understanding, they may lack sector-specific expertise. To solidify our theoretical framework, we delve into the pivotal role of LLMs in modern Natural Language Processing (NLP) research. LLMs, including BERT, Falcon-7B, GPT-2, and MPT-7B, offer substantial improvements over traditional machine learning algorithms by adeptly capturing complex linguistic patterns and contextual nuances within unstructured textual data. Our study innovatively employs prompt engineering techniques to improve the performance of four leading open-source LLMs—FALCON-7B, GPT-2, MPT-7B, and BERT—on consumer feedback, demonstrating their enhanced capacity to capture consumer sentiment and perception. Our research is guided by the following questions:

1. How does prompt engineering technique affect the performance of four Large Language Models (LLMs) in understanding consumer perception?
2. How can falcon-7B based topic modeling techniques be leveraged to identify and categorize consumer perceptions across service-based industries?

This study aims to enhance our understanding of the effectiveness of Large Language Models (LLMs) in interpreting consumer perceptions through detailed experimentation and analysis. The findings from this study will offer valuable insights for strategic decision-making and policy formulation in the service industry. To our knowledge, our study is the first effort to specifically fine-tune LLMs for consumer data, aiming to extract insights from textual data.

2. Literature review

This literature review section aims to provide a comprehensive overview of the evolution of sentiment analysis methods, beginning with traditional rule-based approaches and advancing through machine learning models to state-of-the-art transformer-based language models. We will explore the strengths and limitations of each approach, highlighting the challenges encountered and the strategies employed to address them. Furthermore, the review will delve into the concept of

prompt engineering and its role in enhancing the performance of large language models for domain-specific tasks, such as consumer review analysis.

2.1. Online reviews

Online reviews have become an essential factor in consumer decision-making processes (Mudambi and Schuff, 2010). Research indicates that the likelihood of purchasing a product often increases following recommendations from friends or family, highlighting the influence of the recommendation's source (Djafarova and Rushworth, 2017). There is a growing trend of consumers relying on the experiences of others, with many turning to online reviews as a key information source before finalizing their purchases (O'Connor, 2010; Kim, 2023). In the hospitality sector, the significance of accommodation reviews extends to shaping a hotel's online reputation, influencing its sales, and impacting future earnings (Ögüt and Onur Taş, 2012; Torres et al., 2015). Currently, research on online reviews is primarily focused on sentiment analysis, which involves examining people's emotional responses towards specific products (Stanca et al., 2023). This analysis encompasses identifying feelings of happiness, anger, sadness, joy, criticism, and praise within the reviews, thereby gauging the product's popularity (Liu and Zhang, 2012).

2.2. Sentiment analysis

In the early stages of analyzing consumer reviews, researchers often relied on dictionary-based methods, where they categorized words, phrases, or sentences based on predefined dictionaries (Zhang et al., 2018). These methods were limited in precision, largely because they depended on the quality of the sentiment lexicons and the rules for sentiment analysis. As data volumes expanded, these approaches struggled to cope with the increasingly complex demands of text sentiment classification. In natural language processing, rule-based algorithms often used established lexicon dictionaries like VADER, TEXTBLOG, AFINN, NRC, and BING (Ainapure et al., 2023). However, these methods faced challenges, particularly in capturing the nuanced details of language (Grégoire et al., 2014). They assigned sentiment scores to individual words without considering context, leading to misunderstandings, as seen in the mixed review: "The hotel room was just okay, but the view from the balcony was breathtaking!" (Liu, 2017).

In the given review, the term 'just okay' might be deemed neutral or slightly negative by a sentiment analysis tool, yet the review's overall sentiment leans positive due to the enthusiastic praise of the view. As sentiment analysis incorporates domain-specific jargon, rule-based methods face challenges in recognizing nuanced expressions, limiting their effectiveness (Poria et al., 2020). Moreover, the burgeoning volume of online material poses scalability issues, necessitating regular updates to the lexicons and potentially straining computational resources (Thelwall et al., 2010). The reliance on sentiment dictionaries requires considerable manual updating, a process that becomes increasingly laborious and time-consuming with the vast expanse of online content to analyze (Taboada et al., 2011).

The evolution of machine learning has paved the way for the broader application of text analysis algorithms, such as Random Forests (RF), XGBoost, Naïve Bayes (NB), and decision trees. These methods identify statistical patterns in labeled data, which are then applied to predict or classify new, unseen text (Pang et al., 2002). While they offer improved accuracy compared to dictionary-based approaches, these algorithms often fail to fully grasp the complex semantic connections and contextual nuances present in language. Consider the example: "Although the hotel lobby was clean and modern, the staff seemed indifferent and unhelpful, which did a disservice to the entire experience." Here, the sentiment regarding the hotel is mixed; the positive description of the lobby contrasts with the negative feedback about the staff. The contrast between positive and negative sentiments within a single review

highlights the challenge faced by machine learning models when they depend solely on basic features. Such models may struggle to capture the nuanced complexities of sentiment within context.

The landscape of natural language processing (NLP) has transformed dramatically with advances in deep learning neural networks and training methods. Initially, Recurrent Neural Networks (RNNs) were utilized to process sequential data like text. [Cho et al. \(2014\)](#) explain how RNNs process input sequences in a stepwise manner, leveraging information from prior steps to make predictions. However, RNNs encounter challenges like vanishing gradients, which impair their capacity to maintain long-range dependencies in text data. The concept of vanishing gradients refers to the diminished impact of gradients in the initial layers of RNNs during training, hindering their ability to preserve and communicate information over long text sequences.

To address the limitations of RNNs, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks were developed. [Hochreiter and Schmidhuber \(1997\)](#) emphasized that LSTM models, equipped with memory cells and gating mechanisms, excel at capturing long-range dependencies and overcoming the problem of vanishing gradients observed in conventional RNNs. The design of LSTM allows it to effectively handle long-range data, leading to improved computational efficiency. However, challenges with LSTMs include difficulties in interpretability, as it can be complex to understand the model's decision-making process, and a high sensitivity to hyperparameter settings. Bidirectional LSTMs (BiLSTMs) enhance this capability by analyzing sequences in both forward and reverse directions, thereby improving the model's grasp of context. However, the sequential processing nature of BiLSTMs can result in computational inefficiencies, especially with longer sequences. On the other hand, later introduced encoder-decoder architectures, while advantageous for tasks like machine translation and text summarization, often encounter challenges in maintaining contextual information across lengthy sequences.

A significant breakthrough in NLP was achieved with the introduction of transformer architectures, as detailed in "Attention is All You Need" by [Vaswani et al. \(2017\)](#). Transformers address the limitations of previous models by employing self-attention mechanisms, which enable them to effectively grasp the broader relationships in input sequences. Transformers differ from recurrent architectures in that they process input data in parallel instead of sequentially as seen as [Fig. 3](#). This parallel processing capability makes transformers highly scalable and efficient in handling long sequences. Moreover, transformers are not affected by the vanishing gradient problem due to their non-reliance on sequential processing. [Devlin et al. \(2018\)](#) introduced BERT (Bidirectional Encoder Representations from Transformers), marking a significant progression in transformer-based NLP models. Subsequent research efforts have been aimed at refining and extending BERT, leading to the development of variants such as RoBERTa ([Liu et al., 2019](#)) and DistilBERT ([Sanh et al., 2019](#)). Each of these versions aims to tackle specific issues or incorporate new strategies to enhance the performance and efficiency of pre-trained language models.

The architecture of BERT consists of multiple encoder layers stacked together, with each layer containing self-attention mechanisms and feed-forward neural networks. This structure enabled BERT to revolutionize NLP by delivering state-of-the-art results across various benchmark tasks, attributed to its capability to derive bidirectional context from the text. During preprocessing, BERT's tokenizer converts input text into three types of embeddings: token, segment, and positional. [Devlin et al. \(2018\)](#) describe how token embeddings represent individual words in a sentence as vectors, segment embeddings distinguish different sentences or segments, and positional embeddings provide information about the word positions within the sequence, helping BERT to understand the order of words.

Efforts in research have focused on enhancing the adaptability of pre-trained language models for specific tasks. The paper "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer" by [Raffel et al. \(2020\)](#) examines various fine-tuning strategies and their

effectiveness across different tasks. Furthermore, research on domain adaptation and zero-shot learning ([Xia et al., 2020](#)) aims to enhance the generalizability and applicability of models across new domains, requiring minimal retraining.

As pre-trained language models become larger and more complex, the importance of efficient training and inference methods becomes increasingly evident. To reduce the use of computational resources and improve scalability, researchers are exploring strategies like model compression, parameter sharing, and hardware optimizations ([Goodfellow et al., 2018](#)). The paper "Efficient Transformers: A Survey" by [Tay et al. \(2022\)](#) provides comprehensive reviews of techniques aimed at enhancing the efficiency of language models and Transformer architectures. These advancements are not only pivotal in natural language processing but also play a significant role in computer vision and reinforcement learning.

Transformer-based Large Language Models (LLMs), such as GPT-2 and derivatives like Falcon, have revolutionized text generation and comprehension. By leveraging vast amounts of text data, these models excel at capturing complex language structures and semantic relationships. Prompt engineering has emerged as a strategy to enhance the performance of Large Language Models (LLMs) by crafting tailored prompts for specific tasks or domains ([Wahidur et al., 2024](#)). As explained by [Brown et al. \(2020\)](#), prompt engineering involves crafting prompts that guide the model toward generating desired outputs or performing specific tasks. Furthermore, tuning pre-trained LLMs with domain-specific data enhances their efficacy and relevance for particular tasks or domains. The advent of open-source Large Language Models (LLMs) such as GPT-2, Falcon-7B, and BERT has democratized access to sophisticated Natural Language Processing (NLP) models within the research and development community. This accessibility empowers researchers and developers to leverage cutting-edge techniques without the need for extensive computational resources.

In this study, we selected four open-source Large Language Models (LLMs) - MPT-7B, GPT-2, BERT, for fine-tuning with consumer data. MPT-7B is a versatile LLM capable of handling a wide array of natural language processing tasks. GPT-2 (Generative Pre-trained Transformer 2) is a well-known LLM developed by OpenAI, renowned for its impressive text generation abilities. On the other hand, Falcon-7B is a newer LLM that has demonstrated promising performance across various text analysis tasks. Using prompt engineering techniques, our aim was to enhance the performance of these LLMs in understanding consumer sentiment and perception. Additionally, we aimed to evaluate their performance metrics, with a specific focus on their effectiveness within a particular context.

3. Research methodology

3.1. Data collection and data preprocessing

In this study, our focus was on analyzing consumer reviews due to their significant influence on people's decisions, particularly in the competitive hospitality industry. The sample used in our study is representative, sufficiently large, and unbiased, meeting the assumptions underlying our methodology. We collected consumer reviews from TripAdvisor, a widely used platform for consumer feedback, specifically selecting data from January 2019 to December 2023. This timeframe ensured a diverse range of opinions and experiences from a broad demographic of consumers in the hospitality sector. Our dataset comprises 1,031,478 reviews, evenly split between positive and negative sentiments (515,739 positive and 515,739 negative reviews), ensuring a balanced representation of sentiments. To extract consumer reviews from TripAdvisor's website, we developed a custom Python crawler. Subsequently, we utilized Python libraries to remove duplicate text and special characters. Stop words, including articles and prepositions, were eliminated as they lack inherent significance or value for analysis. After removing stop words, we performed stemming and lemmatization to

standardize words and reduce vocabulary size, thereby enhancing the efficiency of our analysis. Master-level students, recognized for their proficiency in English, annotated the dataset by categorizing reviews as positive or negative.

Our methodology involved fine-tuning the models on the consumer review data specifically for sentiment analysis tasks. We employed advanced Natural Language Processing (NLP) techniques and Large Language Models (LLMs) to analyze a large dataset of consumer reviews from TripAdvisor. Our choice of LLMs is grounded in their proven effectiveness in handling unstructured textual data, as evidenced by prior research. Previous studies have demonstrated the superiority of LLMs over traditional machine learning models in capturing semantic nuances and contextual relationships within text. For instance, Devlin et al. (2018) highlighted the effectiveness of BERT in various NLP tasks, including sentiment analysis and topic modeling. Similarly, research by Radford et al. (2019) on GPT-2 and subsequent work on Falcon-7B (Almazrouei et al., 2023) have shown these models' capabilities in understanding and generating coherent, contextually relevant text. The use of prompt engineering techniques to fine-tune these models further enhances their ability to extract meaningful insights from consumer reviews. Prompt engineering allows us to tailor the models' responses to specific queries, optimizing their performance for sentiment classification and topic extraction. This method is supported by Brown et al. (2020), who demonstrated the effectiveness of prompt engineering in improving model accuracy and relevance. By leveraging these advanced techniques, our study ensures a robust and comprehensive analysis of consumer sentiment and thematic elements within the review data, directly addressing our research questions and providing valuable insights into consumer perceptions and preferences in the hospitality sector.

3.2. Large language models

In recent years, the field of natural language processing (NLP) has undergone a significant transformation due to the rapid development of Large Language Models (LLMs) such as GPT-2, Flan-UL2, LLaMA, MPT, and Falcon. These models undergo rigorous pre-training on vast text corpora, employing techniques like instruction tuning and reinforcement learning with human feedback. The results are remarkable, particularly in scenarios requiring learning from limited examples, leading to a shift from traditional fine-tuning to targeted prompting. Recent studies by Barman et al. (2024) and Wang, W. et al. (2024) showcase the capabilities of LLMs in addressing complex challenges such as polarity shifts, open-domain scenarios, and sentiment inference, highlighting their versatility. 'Perplexity' stands out as a primary metric for evaluating LLMs, quantifying their predictive accuracy as the exponential of the average negative log likelihood per token. This metric reflects the models' proficiency in assigning likelihoods based on context, serving as a crucial measure of their overall performance:

$$\log(\text{Perplexity}) = -\frac{1}{N} \sum_{i=1}^N \log(P(\text{token}_i | \text{context for token}_i))$$

Here, N denotes the number of tokens in the text corpus, and the determination of "context for token_i " hinges on the specific type of LLM in use. For autoregressive LLMs, the context is the text segment preceding token_i , while for masked LLMs, it is the text segment surrounding token_i . Evaluation of Large Language Models (LLMs) typically involves examining their perplexity on a distinct test set, comprising previously unseen data. This practice helps mitigate potential overfitting issues that may arise from training these models on extensive web-scraped corpora. As LLMs continue to evolve, a comprehensive understanding of their performance metrics, ethical considerations, and environmental impacts becomes paramount for their responsible integration into a diverse range of Natural Language Processing (NLP) applications. In this paper, we delve into three different open-source

LLMs—GPT-2, MPT-7B, and Falcon-7B—details of which are elaborated in the upcoming subsections.

3.3. Prompt engineering

Prompt engineering involves creating specific text instructions or cues to guide generative AI systems, such as large language models (LLMs) like GPT-4, in performing particular tasks. Unlike traditional full fine-tuning methods that directly adjust model parameters, prompt engineering focuses on refining prompts to effectively steer the model's output, as illustrated in Fig. 1. This approach leverages the inherent capabilities of LLMs, enabling efficient task adaptation without extensive parameter updates. It contrasts with methods like LoRA (Low-Rank Adaptation) (Hu et al., 2021) or PEFT (Parameter-Efficient Fine-Tuning), as illustrated in Fig. 2. The key advantages of prompt engineering lie in its resource efficiency, time effectiveness, and reduced complexity compared to fine-tuning techniques. Operating solely on user input, it significantly reduces computational and memory resources, making it suitable for resource-constrained environments. Additionally, it streamlines deployment and iteration cycles, enabling quicker adaptation and experimentation with different task configurations. Furthermore, its simplicity at the input level makes it user-friendly for practitioners without extensive deep learning knowledge, providing a straightforward approach to customizing and optimizing tasks. In our study, we utilized prompt engineering techniques to fine-tune four prominent open-source LLMs—BERT, GPT-2, MPT-7B, and Falcon-7B—using a dataset of 1,031,478 consumer reviews. By leveraging vast amounts of pre-trained data, these models were adapted to understand and interpret consumer review data within the hospitality sector. Fine-tuning on domain-specific data enhances their ability to accurately interpret word nuances and semantic meanings, leading to improved performance. Through this innovative approach, we aimed to demonstrate the enhanced capacity of fine-tuned LLMs in capturing consumer sentiment and perception, thereby providing valuable insights for strategic decision-making in the service industry.

We used various prompts to evaluate the performance metrics of the models and made numerous modifications to the prompts to achieve the highest possible metrics. This iterative approach involved experimenting with different types of prompts and incorporating one-shot and few-shot learning techniques to enhance the models' ability to understand and generate accurate responses based on limited examples. Performance metrics such as accuracy, precision, recall, and F1-score were used to evaluate and refine the models, ensuring their suitability for the task at hand.

3.4. Bidirectional Encoder Representation from transformers (BERT)

The Bidirectional Encoder Representations from Transformers (BERT) is a pre-trained language model that operates on the Transformer architecture (Devlin et al., 2018). BERT has demonstrated exceptional performance in natural language processing tasks, particularly in sentiment analysis. On the other hand, Embeddings from Language Models (ELMo) provide distinct representations for each word in an input sequence by combining intermediate layer representations from pre-trained bidirectional long short-term memory networks (See Fig. 4). These representations are then integrated into existing supervised models, such as GloVe, for subsequent tasks. While ELMo significantly enhances various natural language processing tasks, it requires a specific architecture tailored to each task, which can be challenging to implement for every NLP task (Peng et al., 2019). In contrast, the Generative Pre-Training (GPT) model offers a comprehensive framework that is not limited to specific tasks and allows for contextually sensitive representation (Radford et al., 2018). GPT employs the transformer decoder and a pre-trained language model to encode and represent text sequences. During downstream tasks, the pre-trained model's output is fed into an additional linear output layer to make

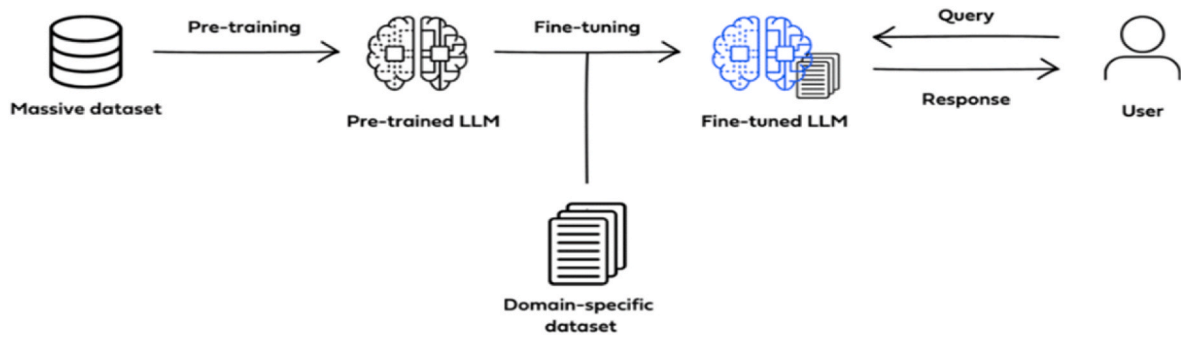


Fig. 1. Full fine tuning.

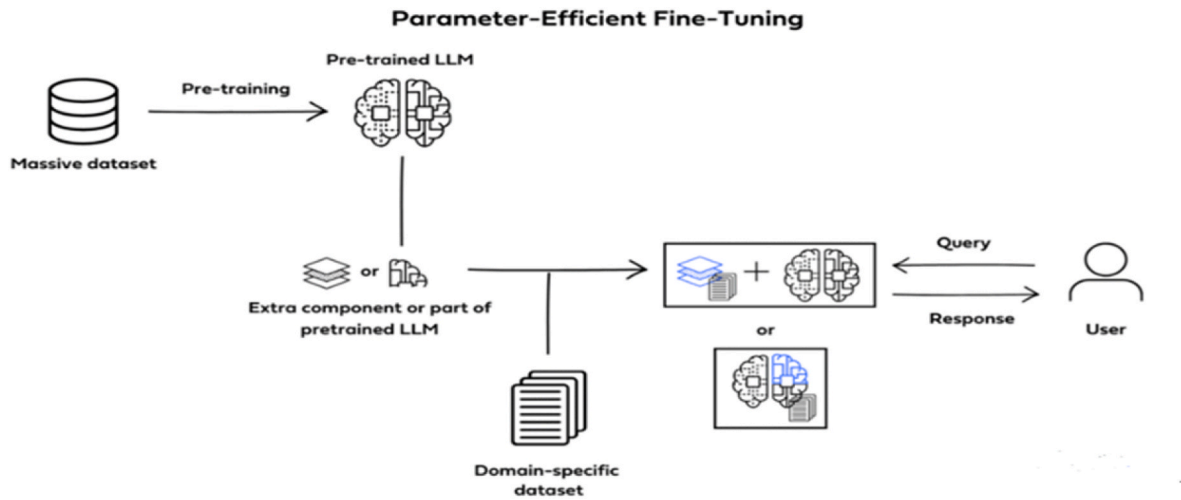


Fig. 2. Parameter- efficient fine-tuning.

predictions for the task label. Unlike ELMo, which maintains fixed parameters of the pre-trained model, GPT adjusts all parameters in the pre-trained Transformer decoder while performing supervised learning on downstream tasks (Goldberg, 2019). ELMo utilizes bidirectional encoding but task-specific architecture, whereas GPT is task-independent but encodes context from left to right (Ethayarajh, 2019). Similarly, BERT, also known as Bidirectional Encoder Representation from Transformers, is effective in various NLP tasks due to its minimal architectural modifications and the use of a pre-trained transformer encoder to represent tokens based on bidirectional context (Devlin et al., 2018).

3.5. Technical concepts

3.5.1. Transformer architecture

The Transformer is a groundbreaking neural network architecture introduced by Vaswani et al. (2017) that incorporates the attention mechanism. This architecture comprises two fundamental components:

3.5.2. Transformer's self-attention mechanism

The self-attention mechanism enables the model to simultaneously consider all positions within the input sequence and compute weights based on their interrelationships. This mechanism is mathematically represented by the following formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Here, Q , K , and V are the query, key, and value matrices, respectively, and d_k is the dimensionality of the critical vectors.

3.5.3. BERT's bidirectional encoder

BERT's bidirectional encoder primarily consists of the following components:

- **Multi-head Self-Attention Encoder:** This component utilizes multiple self-attention heads (Multi-head Attention) to capture context relationships at different levels within the input sequence (Vaswani et al., 2017).
- **Positional Encoding:** To retain positional information in the Transformer sequence, positional encoding employs a one-dimensional periodic sin/cos function encoding scheme.
- **Layer Normalization:** This normalization technique is crucial for mitigating gradient-related issues such as gradient explosion and vanishing gradients, ensuring stable training of the model (Xiong et al., 2020).

3.5.4. BERT's pre-training tasks

BERT's pre-training tasks primarily comprise the following two tasks:

- **Masked Language Modeling (MLM):** This task requires BERT to predict the context of masked words. This task can help the model understand the relationship between words in context.
- **Next Sentence Prediction (NSP):** In this task, BERT must predict the relationship between a sentence and its preceding sentence. This task facilitates the model's acquisition of the correlation between sentences.

BERT model with an encoder-style architecture designed for

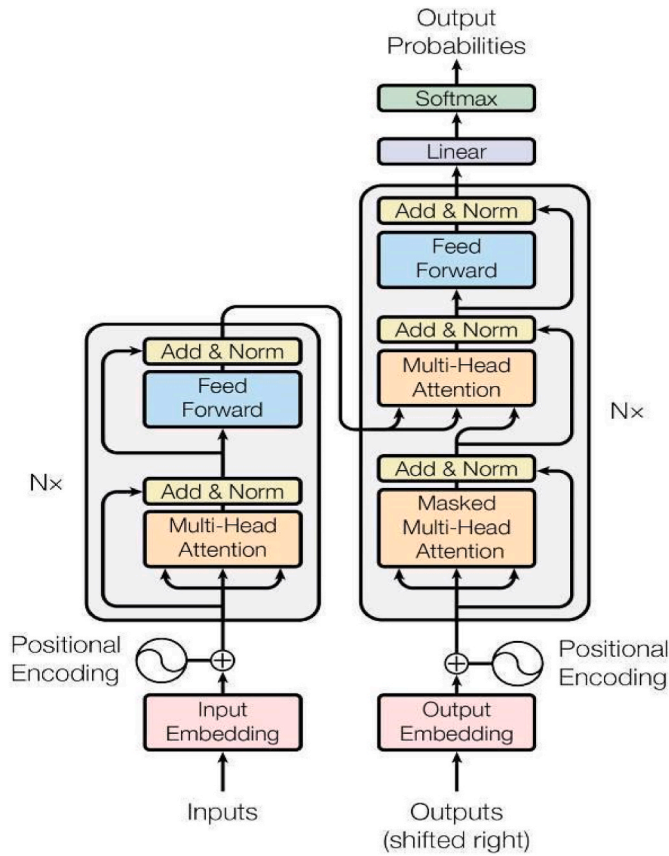


Fig. 3. Transformer model architecture.
Sources: Vaswani et al. (2017).

predictive modeling tasks (Devlin et al., 2018).

3.6. GPT-2

Generative Pre-trained Transformer 2 (GPT-2), unveiled in November 2019, marks a significant leap forward in OpenAI's GPT series, succeeding GPT-1 and paving the way for subsequent iterations

such as GPT-3 and GPT-4 (Radford et al., 2019). Boasting a ten-fold increase in parameters and training data compared to its predecessor, GPT-2 underwent training on a diverse corpus comprising over 7000 fiction books from BookCorpus and a staggering 8 million web pages from WebText. Its versatility is evident as a general-purpose learner, showcasing excellence in tasks like text translation, question-answering, and summarization. Despite its impressive capabilities, GPT-2 does have occasional limitations, particularly in generating repetitive or nonsensical content in longer texts. Based on the transformer architecture shared with other GPT models, GPT-2 utilizes attention mechanisms to focus on specific input segments, allowing for efficient parallel processing (Fig. 5). While GPT-2 played a significant role in advancing language models, newer iterations have surpassed its capabilities. The transformer architecture used in GPT-2 has become fundamental in natural language processing, advancing innovation with attention mechanisms and improved parallel processing capabilities. Unlike earlier recurrence- and convolution-based architectures, the transformer model's use of attention mechanisms enables selective focus on important parts of input text, resulting in significant improvements in parallelization and surpassing previous benchmarks set by RNN/CNN/LSTM-based models.

GPT-2 is fundamentally an autoregressive model. It predicts the next word in a sequence based on the words that came before it. This prediction process iterates until the desired length of text is generated. GPT-2 employs a softmax function to calculate the probability distribution across the vocabulary for every word in the sequence.

3.7. MPT-7B

MPT-7B, part of the Mosaic Pretrained Transformer (MPT) series, stands as a decoder-style transformer model that underwent comprehensive pretraining from scratch on an extensive dataset comprising 1 trillion tokens of English text and code. Developed by MosaicML, MPT-7B features a customized transformer architecture designed specifically for efficient training and inference processes. This model incorporates significant adaptations, including optimized layer implementations aimed at enhancing performance and the adoption of Attention with Linear Biases (ALiBi) to replace positional embeddings, effectively eliminating context length limitations. These architectural refinements are pivotal in empowering MPT models like MPT-7B to achieve exceptional throughput efficiency during training while ensuring stable

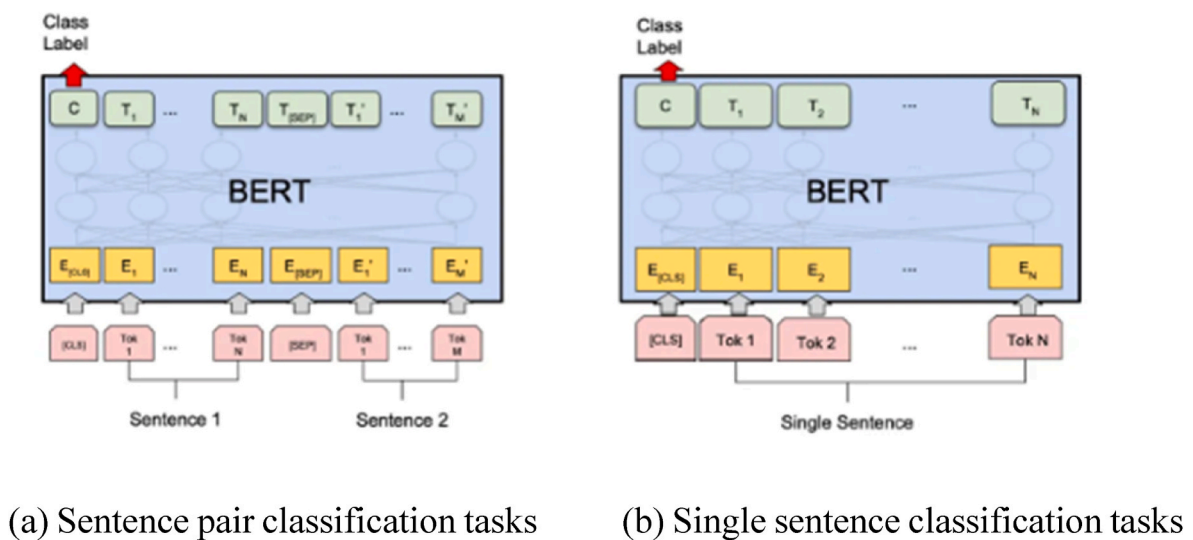


Fig. 4. BERT model architecture for sentence-pair and single-sentence classification.
Sources: Devlin et al. (2018).

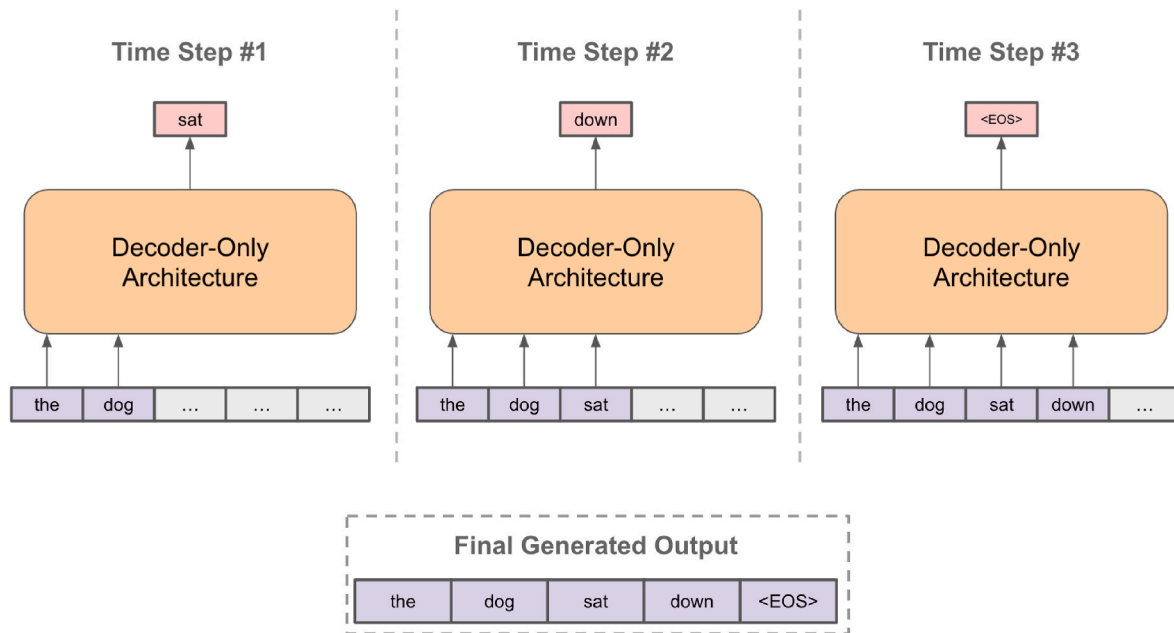


Fig. 5. Text generation process using decoder-only architecture.

Sources: Radford et al. (2019).

convergence. The contributions made by the MosaicML NLP Team in refining these architectural elements have resulted in a model that excels in processing large volumes of data with efficiency and reliability, making MPT-7B a valuable asset in natural language processing tasks.

3.8. Falcon-7B

Falcon, a new advancement in language modeling created by the Technology Innovation Institute in Abu Dhabi, includes a set of state-of-the-art language models released under the Apache 2.0 license. Notably, Falcon-7B and Falcon-40 B are significant additions to modern language models, focusing on inference efficiency by training on extensive datasets with 1.5 trillion and 1 trillion tokens, respectively. The exceptional quality of the Falcon models originates from their training data, primarily sourced from RefinedWeb, a cutting-edge web dataset built from CommonCrawl. This dataset surpasses traditional curation methods by prioritizing scalability and data quality enhancement through thorough deduplication and meticulous filtering processes. This rigorous approach strives to meet or exceed the quality standards set by other well-known corpora.

While the Falcon models incorporate curated sources like conversational data from platforms such as Reddit, they rely less on such sources compared to leading language models. One standout feature of the Falcon models is the introduction of multi-query attention, a novel adaptation of the attention mechanism in neural networks (Fig. 6). In traditional attention mechanisms, each attention head has its query, key, and value components. However, the Falcon models introduce multi-query attention, where a single key and value are shared across all attention heads, presenting a unique approach to optimizing attention mechanisms (Almazrouei et al., 2023). Mathematically, the concept of multi-query attention can be represented as follows:

$$\text{MultiQueryAttention}(Q, K, V) = \text{softmax}\left(\frac{Q(K_{\text{shared}})^T}{\sqrt{d_k}}\right)V_{\text{shared}}$$

Here Q represents queries, K_{shared} is the shared key, V denotes values, and V_{shared} represents the shared value. This modification enhances the models' ability to attend to information uniquely and efficiently.

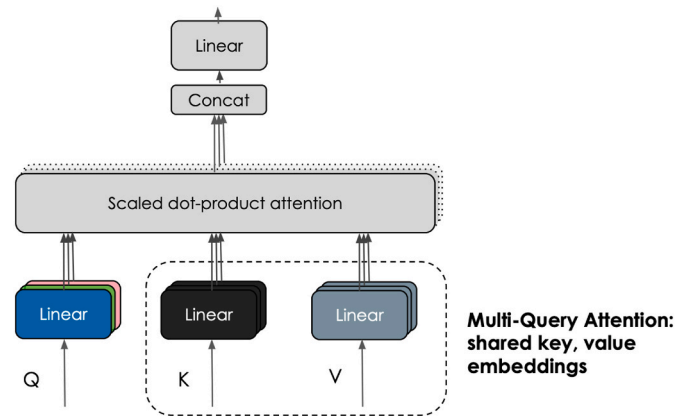


Fig. 6. Multi-query attention mechanism with shared key and value embeddings.

Sources: Almazrouei et al. (2023).

3.9. Topic modeling

The abundance of digital content on the Internet, including news articles, social media discussions, and scholarly publications, has posed a challenge for researchers to extract valuable insights from large datasets efficiently. Manual content analysis, a conventional method for extracting information from online reviews, faces obstacles due to labor-intensive processes and scalability limitations (Krippendorff, 2018). Researchers often struggle to identify relevant attributes or summarize paragraphs that capture the core of these extensive datasets. To tackle these challenges, topic modeling has become a valuable technique for uncovering latent themes within a corpus. Essentially, topic modeling is a statistical method that uses unsupervised learning to group the hidden semantic structure of a corpus. This method helps address issues like polysemy and ambiguity in topic discovery by utilizing traditional clustering algorithms and considering language complexity more fairly, requiring deeper contextual analysis and adaptability (Rajput et al., 2020). Latent semantic indexing (LSI) offers a solution to some of these challenges but introduces the potential for semantic inaccuracies

(Papadimitriou et al., 1998; Xu and Li, 2016). Latent Dirichlet Allocation (LDA), introduced by Blei et al. (2003), addresses these concerns by estimating topic distributions within a corpus based on the co-occurrence probability between words. An extension of LDA known as Structural Topic Modeling (STM) enhances the comprehension of topic relationships. (Liu et al., 2022; Sánchez-Franco et al., 2021). Despite its advantages, STM requires further refinement to achieve ideal attributes (Sv and Vijaya, 2024).

Advanced techniques like deep learning transformer models, particularly pre-trained BERT models, play a significant role in text analysis. These models leverage deep learning and attention mechanisms to capture complex language structures and semantic nuances within sentences. BERTopic, a topic modeling approach utilizing BERT, demonstrates the potential of combining state-of-the-art language models with topic extraction clustering algorithms (Grootendorst, 2022; Praveen and Deepika, 2023; Praveen and Vijaya, 2023). The utilization of transformer-based architectures such as Falcon-7B for topic modeling through prompts offers notable advantages in terms of enhanced contextual understanding, flexibility, and performance (Almazrouei et al., 2023). By harnessing the capabilities of these advanced models, researchers can extract deeper insights from textual data, facilitating informed decision-making and knowledge discovery across various domains, as demonstrated in our study.

4. Results and discussions

In this study, we employed four state-of-the-art large language models (LLMs) – MPT-7B, GPT-2, Falcon-7B, and BERT – to analyze a massive dataset of over one million consumer reviews from TripAdvisor. By leveraging the computational power of NVIDIA GPUs, known for their capability to handle vast amounts of data, we trained these LLMs on this extensive and diverse dataset. This approach helped mitigate potential biases and ensured a comprehensive representation of consumer opinions and experiences. Utilizing cutting-edge LLMs and a large training dataset, facilitated by high-performance computing resources, yielded impressive performance metrics. This allowed us to overcome the limitations of rule-based algorithms, traditional machine learning models, and lower-capacity LLMs with fewer parameters, which often result in less accurate performance metrics.

4.1. Performance metrics for sentiment analysis

Metrics play a crucial role in evaluating performance and effectiveness across diverse domains, including hospitality, tourism, healthcare, and service-based industries. These quantifiable measures offer valuable insights that aid stakeholders in making informed decisions, identifying areas for enhancement, and monitoring the progress of initiatives or strategies.

4.1.1. Key metrics and formulas

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{F1 score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Samples}}$$

Precision, recall, F1 score, and accuracy score are fundamental metrics for assessing the performance of classification models. Each metric offers unique insights into model effectiveness and aids in comprehending the balance between false positives and false negatives.

Integrating these metrics provides data scientists with a holistic view, enabling informed decisions regarding model selection, parameter tuning, and feature engineering to enhance overall model performance. Table 1 compares the performance metrics (F1 score, accuracy, precision, and recall) for the four language models (MPT-7B, GPT-2, Falcon, and BERT) to compare their task effectiveness.

The F1 score, derived from the harmonic mean of precision and recall, offers a balanced assessment of model performance in terms of both precision and recall. A higher F1 score signifies superior performance in achieving this balance. According to F1 scores, Falcon leads with the highest score of 0.91348, showcasing its effectiveness in striking a balance between precision and recall. However, it's crucial to note that accuracy, while significant, might not provide the most accurate evaluation for imbalanced datasets.

Falcon-7B achieved the highest accuracy scores of 0.914, highlighting its overall ability to predict instances accurately. Precision, which measures the ratio of correctly predicted positive observations to total predicted positive observations, indicates how well the model identifies positive instances. Falcon-7B exhibits the highest precision score of 0.91906, closely followed by MPT-7B at 0.89181.

Recall, also referred to as sensitivity, assesses how well the model captures actual positive instances. Falcon-7B achieves the highest recall score of 0.914. Considering these metrics collectively, Falcon-7B emerges as the top-performing model among the four, showcasing superior performance in terms of F1 score, accuracy, precision, and recall.

4.2. Topic modeling analysis

The comparative analysis between Latent Dirichlet Allocation (LDA) and Falcon-7B LLM offers valuable insights into their respective capabilities in topic modeling. LDA employs a probabilistic Bayesian model and word distributions to uncover latent topics in textual data, while Falcon-7B LLM leverages transformer-based architectures and deep learning techniques to capture semantic relationships. Our topic modeling results are shown in the Tables 2 and 3.

In Table 2, which presents the results from LDA, a notable lack of thematic coherence is evident in the identified topics. For instance, terms like 'printer' in 'Food Service', 'lamp' in 'Staff Attitude', 'didn't' in 'Complaint service', 'cancel' in 'Neighborhood', and 'perfect' in 'Bathroom access' seem randomly included based solely on word frequency rather than a nuanced understanding of underlying concepts. In the 'Bathroom Access' topic, while relevant terms such as 'sink,' 'heater,' 'towel,' and 'tap' contribute to discussions on facilities and amenities, the inclusion of 'disappoint' and 'wet' appears disconnected and lacks contextual relevance. Similarly, the term 'perfect' in the same category stands out as an outlier, seemingly inserted without a clear connection to the discussion on bathroom features. These outcomes are consistent with expectations from LDA's Bayesian probabilistic approach, which relies on the likelihood of observing specific words within topics.

Conversely, Table 3, displaying the outcomes from Falcon-7B based topic modeling, showcases well-defined and intuitively sensible topics. These coherent topics suggest that Falcon LLM, likely owing to its transformer architecture, can discern relationships between words and generate semantically cohesive topics. Furthermore, Falcon-7B LLM demonstrated superior performance in topics requiring contextual understanding and semantic coherence, such as Front Desk and Overall Atmosphere. In these topics, Falcon-7B LLM exhibited a higher degree of

Table 1
Comparison performance metrics.

Model	F1 Score	Accuracy	Precision	Recall
MPT-7B	0.88759	0.888	0.89181	0.888
GPT-2	0.5075	0.5647	0.6207	0.56467
Falcon-7B	0.91348	0.914	0.91906	0.914
BERT	0.7266	0.73667	0.77727	0.73667

Table 2
LDA based Topic modeling.

Topic Label	Top Words
Food Service	Egg, breakfast, bacon, toast, printer, pastry, warm
Service Failure	Unhelpful, terrible, chair, attitude, rude, poor
Staff Attitude	Friendly, helpful, generous, delight, lamp, sweet, smile
Lift	Lift, one, minute, crawl, space, block, pencil
Cleanliness	Good, feel, clean, fast, base service, delay
Front Desk	Little, front, desk, morn, stain, cheap, printer
Suite Equipment	Sofa, chair, wardrobe, cramp, unsort, aircon, book
Complaint Service	Complaint, problem, delay, late, struggle, lack, didn't
Decoration	Overdone, frame, pass, silk, style, décor, pillow
Noise	Room, noise, sleep, annion, discuss, horrible
Recreation Facility	Bar, gym, music, pool, kettle, theatre, book
Refund Procedure	Told, found, refund, receive, late, require, pencil
Professionalism	Staff, great, professional, polite, warm, awesome, lamp
Expectation Failure	Hotel, everything, didn't, expense, much, cost, expect
Shower	Sink, heater, towel, disappoint, drain, tap, wet
Bathroom Access	Bathrobe, shampoo, soap, perfect, size, hair dryer, soft
Neighborhood	Parliament, supermarket, cancel, district, subway, soho
Room Convenience	Room, comfort, realli, special, entertain, huge, charm
Overall	Welcome, light, bright, relax, convenient, atmosphere, printer
Atmosphere	Printer
Public Facility	Taxi, corridor, main, danger, don't, elder, delay

Table 3
Falcon-7B based Topic Modeling.

Topic Label	Top Words
Safety Fear	Elevator, parking, stairs, narrow, security, sanitation, hazard
Noise	Distraction, noise, loud, sleep, quiet, peaceful
Food Service	Tea, coffee, eggs, pastries, breakfast, delicious, flavorful
Room Facilities	Pillow, bed, sofa, clean, spacious, wardrobe, comfortable
Staff Attitude	Courteous, helpful, welcoming, attentive, friendly, pleasant, professional
Recreation Facilities	Spa, bar, wine, pool, music, theater, relax
Celebration	Family, weekend, birthday, anniversary, disappointment, party, festive
Bathroom Size	Small, cramped, tiny, shower, toilet, sink, compact
Shower Experience	Shower, water, pressure, temperature, drain, toiletries, refreshing
Decent Location	Convenient, central, accessible, station, supermarket, walkable
Booking Experience	Hotel, reservation, online, confirmation, refund, communication, smooth
Bedroom	Clean, bed, comfortable, quiet, dark, restful, relaxing
Environment	
Transportation	Public, taxi, bus, train, metro, accessible, navigate
Bathroom Access	Soap, shampoo, conditioner, towels, robe, toiletries, luxurious
Housekeeping	Housekeeping, cleaning, towels, linens, tidy, attentive, thorough
Poor Maintenance	Window, door, broken, worn, outdated, repair, maintenance
Complaint Service	Complaint, issue, resolved, responsive, apologetic, accommodating, empathy
Internet	WiFi, connection, fast, reliable, streaming, work
Neighborhood	Welcoming, friendly, charming, special, memorable, delightful, exceed
Good Feeling	Lively, vibrant, restaurants, shops, cafes, safe, walkable

coherence and relevance in the extracted keywords, providing a more comprehensive representation of the underlying themes.

For instance, in the Food Service topic, Falcon-7B LLM accurately identified keywords like “tea,” “coffee,” “eggs,” and “pastries,” highlighting essential elements for a satisfying breakfast experience. It also emphasized descriptors such as “delicious” and “flavorful,” reflecting the quality and taste of the food, thus enhancing overall guest satisfaction. Regarding Room Facilities, Falcon-7B LLM adeptly recognized keywords like “pillow,” “bed,” “sofa,” and “clean,” underscoring crucial aspects of comfort and hygiene. Furthermore, it identified descriptors like “spacious” and “wardrobe,” emphasizing the importance of ample

space and storage facilities, contributing to a pleasant stay for guests. In the domain of Staff Attitude, Falcon-7B LLM accurately identified keywords such as “courteous,” “helpful,” “welcoming,” and “attentive,” reflecting attributes crucial for positive guest-staff interactions. Moreover, it highlighted terms like “friendly,” “pleasant,” and “professional,” indicating the expected demeanor and conduct from hospitality personnel, thereby enriching the overall guest experience.

Similarly, in topics like ‘Safety Fear,’ which groups words like ‘security,’ ‘sanitation,’ and ‘hazard,’ and ‘Shower Experience,’ linking words like ‘water pressure,’ ‘temperature,’ and ‘refreshing,’ Falcon-7B LLM showcased a depth of understanding in sentiment and tone. For instance, in the Staff Attitude topic, Falcon-7B LLM highlighted words like “friendly,” “helpful,” and “generous,” indicating a positive sentiment towards staff behavior. Conversely, in the Service Failure topic, it captured words like “attitude” and “rude,” providing insights into negative service experiences.

By influencing contextual understanding and semantic coherence, Falcon LLM transcends the limitations of traditional bag-of-words approaches like LDA, allowing for a more insightful interpretation of underlying themes within textual data.

5. Conclusion

In this research, we investigated the capabilities of state-of-the-art Large Language Models (LLMs) such as BERT, MPT-7B, GPT-2, and Falcon-7B for analyzing consumer reviews. Our focus was on sentiment analysis and topic modeling, utilizing advanced Natural Language Processing (NLP) techniques to process a substantial volume of unstructured consumer feedback sourced from the TripAdvisor platform. The study revealed that LLMs, particularly Falcon-7B, excel at capturing complex linguistic patterns and contextual nuances, outperforming traditional machine learning algorithms and basic NLP methods in understanding consumer data. Precision, recall, and F1 scores for Falcon-7B underscored its superior performance in interpreting consumer sentiments. This model’s efficiency and capability to capture subtle language nuances are crucial for businesses aiming to deeply understand customer feedback. Our study contributes significantly to the field of NLP and consumer review analysis. By leveraging advanced features of LLMs, such as self-attention mechanisms and transfer learning, we demonstrated their superiority over conventional methods like rule-based algorithms and Latent Dirichlet Allocation (LDA) in both sentiment analysis and topic modeling tasks. The advanced capabilities of LLMs in understanding complex language patterns and contextual subtleties lead to more accurate and insightful interpretations of consumer reviews. Moreover, the study showcases the efficacy of prompt engineering as a cost-efficient alternative to traditional fine-tuning techniques. This method enables rapid adaptation of LLMs to specific business requirements, providing a valuable tool for companies needing to respond swiftly to dynamic market conditions. Highlighting the revolutionary potential of LLMs in extracting actionable business intelligence from consumer feedback, our research underscores their significant impact on strategic decision-making and customer experience strategies. Building on foundational work by Devlin et al. (2018) with BERT and Radford et al. (2019) with GPT-2, we extended their applications to consumer review analysis. The inclusion of Falcon-7B, a state-of-the-art LLM, exemplifies the advancements in NLP techniques. Our comprehensive evaluation and application of these models contribute to a deeper understanding and broader utilization of LLMs, marking a pivotal development in consumer review analysis. Through this innovative approach, we provide valuable insights that enhance current methodologies and demonstrate the transformative potential of LLMs in business intelligence.

This study addresses a topic of substantial applied relevance, particularly in business sectors such as hospitality, tourism, healthcare, and other service industries where understanding customer feedback is crucial for service improvement and customer satisfaction. By utilizing

optimized NLP models, specifically advanced Large Language Models (LLMs) like Falcon-7B and BERT, our research emphasizes the practical importance of analyzing consumer reviews to extract valuable insights. The application of sentiment analysis and topic modeling using these state-of-the-art models provides businesses with detailed information about consumer sentiments and trends, enabling them to make informed decisions. One of the key practical contributions of our study is the demonstration of prompt engineering as a cost-effective method for customizing LLMs to specific industry needs without incurring high computational costs. This approach allows businesses to quickly adapt to changing market dynamics and tailor their strategies to meet customer expectations effectively. By providing a framework for the rapid deployment of LLMs in various industry contexts, our research highlights the real-world applicability of these advanced NLP techniques. The overall findings of our study offer actionable insights for business leaders, helping them to respond swiftly to market changes and optimize customer experience strategies. This applied relevance underscores the urgent need for modern businesses to leverage data-driven approaches in understanding and addressing consumer feedback, thereby enhancing their competitive edge and ensuring customer satisfaction.

Additionally, we utilized prompt engineering as an effective strategy to tailor these models to specific industry needs without the significant computational expense associated with traditional fine-tuning methods. This technique allows for flexible adaptation to new trends or crisis situations, which is vital for businesses needing to respond quickly to changing market dynamics. Our comparative analysis of topic modeling techniques revealed that while Latent Dirichlet Allocation (LDA) can identify basic themes, it lacks the ability to understand deeper contextual relationships between terms. In contrast, Falcon-7B's transformer-based method offers a more dynamic and insightful approach, significantly improving the quality and relevance of insights drawn from consumer reviews. This research underscores the transformative impact of LLMs in extracting valuable business insights from consumer feedback, crucial for strategic decision-making and policy-making in the service industry. The capability of these models to process and analyze vast datasets not only enhances understanding of consumer needs but also empowers businesses to improve their customer experience strategies effectively.

6. Theoretical and managerial implications

The theoretical underpinnings of our study, grounded in the advancements of Large Language Models (LLMs), significantly influenced our findings. LLMs' ability to capture nuanced sentiments and thematic elements in consumer reviews demonstrates their superiority over traditional machine learning methods. This validates our methodological choices and underscores the importance of adopting modern NLP techniques in consumer behaviour research. The effectiveness of LLMs in extracting detailed insights from unstructured textual data highlights their potential to revolutionize the analysis of consumer feedback, providing a deeper and more accurate understanding of consumer sentiments and preferences. The findings from this study have significant implications for businesses across various sectors, aiding them in understanding consumer opinions and sentiments. This understanding is particularly valuable for enhancing customer satisfaction and improving service quality, leading to informed decision-making and strategic planning. Falcon-7B's superior performance in sentiment analysis and topic modeling showcases LLMs' potential to provide actionable insights from large volumes of reviews. Prompt engineering offers a cost-effective alternative, enabling rapid adaptation to industry-specific contexts for informed decision-making. Falcon-7B's prowess in capturing contextual relationships underscores the necessity for businesses to adopt state-of-the-art LLMs for a competitive edge in customer understanding and response. For managers, leveraging LLMs like Falcon-7B can lead to more precise targeting of customer needs and preferences, thereby enhancing customer satisfaction and loyalty. The

ability to swiftly analyze and respond to feedback can improve service quality and operational efficiency. Furthermore, integrating these advanced NLP techniques can foster innovation in customer experience strategies, setting a benchmark for industry standards and driving long-term business success.

6.1. Limitations and future research

The field of large language models (LLMs) and artificial intelligence (AI) is rapidly evolving, with new architectures and techniques continually emerging. Future research could explore the applicability and performance of more recent LLMs, such as LLaMA 3, released in April 2024, and future versions of open-source GPTs. This exploration could potentially uncover unique strengths or limitations in various domains or tasks. Additionally, this study's scope was limited to English language data, leaving room for future research to address the challenges of understanding code-mixed text and under-resourced languages or domains. Such efforts would enhance the generalizability and inclusivity of LLM applications. Moreover, the computational requirements and inference time associated with LLMs remain significant constraints, particularly in real-time or resource-constrained environments. Future studies could focus on accelerating inference time and improving computational efficiency through algorithmic optimizations, hardware acceleration, or model compression techniques. Exploring prominent state-of-the-art techniques, such as Retrieval-Augmented Generation (RAG) models, could also be beneficial. By addressing these areas, future research can contribute to more efficient, inclusive, and versatile applications of LLMs across different sectors and linguistic contexts.

To enhance the robustness and forward-compatibility of our research, we propose adaptable evaluation and scoring criteria for sentiment analysis using LLMs. These criteria are designed to evolve with newer models and focus on key performance metrics such as accuracy, precision, recall, and F1-score. These metrics are universally applicable across different languages and contexts, making our approach scalable and suitable for benchmarking LLMs across diverse linguistic datasets. This ensures that insights from our research remain valuable as the field progresses, providing a standardized framework for future advancements in multilingual sentiment analysis.

This study contributes significantly to the field of natural language processing (NLP) and its application in consumer review analysis. By leveraging state-of-the-art LLMs such as Falcon-7B, we showcase the superior performance of advanced transformer-based models in sentiment analysis and topic modeling. The incorporation of prompt engineering for customizing LLMs to industry-specific needs without extensive fine-tuning represents a methodological advancement, providing a cost-effective and flexible solution for businesses. Our comparative study between Latent Dirichlet Allocation (LDA) and Falcon-7B in topic modeling offers insights into the limitations of traditional techniques and the advantages of transformer-based models. The research emphasizes the practical implications of utilizing Large Language Models (LLMs) across industries, highlighting their potential to offer actionable insights from consumer reviews. This extends to businesses in the service industry and beyond, showcasing the value of LLMs in understanding consumer sentiments and guiding strategic decision-making. The study underscores the significance of advanced NLP techniques in extracting valuable business intelligence, aiding strategic decision-making processes, and enhancing customer experience strategies. Overall, this research contributes to the advancement of NLP by demonstrating the practical and methodological benefits of utilizing state-of-the-art LLMs for consumer review analysis. It offers new perspectives for both academic research and industry practice, paving the way for further developments in the field.

CRedit authorship contribution statement

S.V. Praveen: Writing – review & editing, Investigation, Formal

analysis, Data curation, Conceptualization. **Pranshav Gajjar:** Validation, Software, Project administration, Methodology, Investigation, Formal analysis. **Rajeev Kumar Ray:** Writing – original draft, Visualization, Supervision, Investigation, Conceptualization. **Ashutosh Dutt:** Writing – review & editing, Writing – original draft, Software.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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